Power BI Assignment:

Introduction to Power BI, Charts, DAX & Creating Reports:

1. Define Power BI, and what are the key components of the Power BI ecosystem? Briefly explain: Power BI Desktop, Power BI Service, Power BI Mobile, Power BI Gateway

Microsoft Power BI is a Business Intelligence (BI) and Data Visualization tool developed by Microsoft. It helps organizations collect, analyze, and visualize data to make better business decisions. Power BI converts raw data from different sources (Excel, SQL databases, cloud services, etc.) into interactive dashboards, reports, and charts.

Key Components of the Power BI Ecosystem

1. **Power BI Desktop:-** Power BI Desktop is a free application installed on a computer.

Purpose

- Used to connect to data sources
- Clean and transform data using Power Query
- Create data models
- Build reports and visualizations

Key Features

- Data transformation
- DAX calculations
- Interactive charts
- Report creation

Example:
A data analyst imports Excel sales data, cleans it, and creates a sales dashboard.

2. **Power BI Service:-** Power BI Service is the cloud-based platform where reports and dashboards are published and shared online.

Purpose

- Share reports with team members
- Create dashboards
- Schedule data refresh
- Collaborate with colleagues

Key Features

- Web-based access
- Dashboard sharing
- Real-time updates
- Workspace collaboration

Example:
After creating a report in Desktop, the analyst publishes it to the Power BI Service so managers can view it online.

3. **Power BI Mobile:-** Power BI Mobile is the mobile application for smartphones and tablets.

Purpose

- Access dashboards and reports anytime, anywhere

Key Features

- Mobile-friendly dashboards
- Real-time notifications
- Touch-enabled interaction

Example:
A manager checks daily sales performance on their phone while traveling.

4. **Power BI Gateway:-** Power BI Gateway connects on-premises data sources to the Power BI cloud service.

Purpose

- Securely transfer data from local databases or servers to the Power BI Service.

Key Features

- Automatic data refresh
- Secure data transfer
- Supports multiple data sources

Example:
A company stores data in a local SQL Server. The Gateway sends this data to the Power BI Service so dashboards stay updated.

Question 2: Compare the following Power BI visuals:
● Pie Chart vs Donut Chart ● Bar Chart vs Column Chart When would you prefer one over the other? Give one example for each pair.

Pie Chart vs Donut Chart

Pie Chart

A Pie Chart shows how different categories contribute to a whole (100%).

Features

- Circle divided into slices
- Each slice represents a percentage of the total
- Best for a few categories (3–5)

When to Use

- When you want to show the percentage distribution of one dataset

Example

Sales by product category:

- Electronics – 40%
- Clothing – 30%
- Furniture – 20%
- Others – 10%

A Pie Chart clearly shows which category has the largest share.

Donut Chart

A Donut Chart is similar to a Pie Chart but has a hole in the center.

Features

- Same concept as a pie chart
- Center space can show the total value or the KPI

When to Use

- When you want to display the total value in the center
- When you want a more modern dashboard design

Example
Monthly revenue distribution:

- Online Sales – 60%
- Store Sales – 40%

The center of the donut can show Total Revenue: ₹10M.

Bar Chart vs Column Chart

Bar Chart

A Bar Chart displays data using horizontal bars.

Features

- Categories on the Y-axis
- Values on X-axis
- Works well when category names are long

When to Use

- When you have many categories
- When labels are long

Example
Top 10 products by sales:

- Laptop – ₹500K
- Mobile Phone – ₹450K
- Smart Watch – ₹300K

Horizontal bars make it easier to read long product names.

Column Chart

A Column Chart uses vertical bars.

Features

- Categories on X-axis
- Values on the Y-axis
- Best for time-based comparisons

When to Use

- When comparing values across time periods

Example
Monthly sales:

Month	Sales
Jan	₹20K
Feb	₹25K
Mar	₹30K

A column chart clearly shows the growth trend month by month.

Question 3: Explain the significance of: ● Star schema vs Snowflake schema ● Primary key vs Foreign key in relationships (Power BI). Why is cardinality important?

Star Schema

A Star Schema is a data model where one central fact table connects to multiple dimension tables.

Structure

- Fact Table → contains numerical data (sales, profit, quantity)
- Dimension Tables → contain descriptive information (product, customer, date)

Why is it important

- Simple structure
- Faster query performance
- Easier to understand

Example

Fact Sales Table

- ProductID
- CustomerID
- DateID
- SalesAmount

Dimension Tables

- Product Table
- Customer Table

- Date Table

All dimension tables connect directly to the fact table, forming a star shape.

When to use

- Most Power BI dashboards
- Data analytics and reporting

Snowflake Schema

A Snowflake Schema is a more normalized version of a star schema where dimension tables are split into additional tables.

Structure

- The fact table connects to the dimension tables
- Dimension tables connect to other dimension tables

Example

Product Table
→ Category Table
→ Subcategory Table

Instead of one product dimension, it is divided into multiple related tables.

Advantages

- Reduces data redundancy
- Better for complex databases

Disadvantages

- More joins
- Slower performance
- More complex model

Primary Key vs Foreign Key:

Primary Key

A Primary Key is a unique identifier for each record in a table.

Characteristics

- Unique values
- Cannot contain duplicates
- Identifies each row

Example

Product Table

ProductID	ProductName
101	Laptop
102	Phone

Foreign Key

A Foreign Key is a column that links one table to another table.

It references the Primary Key in another table.

Example

Sales Table

SaleID	ProductID	Amount
1	101	500
2	102	300

Why Cardinality is Important

Cardinality defines how tables relate to each other.

Common types in Microsoft Power BI:

1️⃣ One-to-Many (1:*)
One record in Table A connects to many records in Table B.

Example
1 Product → Many Sales

This is the most common relationship in Power BI.

2️⃣ Many-to-One (*:1)
Many records connect to one record.

Example
Many sales → One product.

3️⃣ Many-to-Many (:)
Multiple records in both tables relate to each other.

Example
Students ↔ Courses

Question 4: Differentiate between: ● Calculated column vs Measure. Also, define Row context and Filter context with simple examples.

In Microsoft Power BI, Calculated Columns, Measures, Row Context, and Filter Context are important concepts used with DAX (Data Analysis Expressions) to analyze data and build powerful reports.

Calculated Column vs Measure

Calculated Column

A Calculated Column is a new column created using a DAX formula in a table.

Characteristics

- Calculated row by row
- Stored in the data model
- Increases file size
- Can be used in slicers, filters, rows, and columns

Example

Sales Table:

Product	Quantity	Price
Laptop	2	500

Phone	3	300

Calculated Column formula:

TotalPrice = Sales[Quantity] * Sales[Price]

Measure

A Measure is a calculation performed on aggregated data.

Characteristics

- Calculated dynamically
- Not stored in the table
- Used mainly in visualizations
- Does not increase model size

Example

Measure formula:

Total Sales = SUM(Sales[TotalPrice])

Row Context

Row Context means the formula operates on one row at a time.

Each row is evaluated separately.

Example using a calculated column:

Profit = Sales[Revenue] - Sales[Cost]

Filter Context

Filter Context refers to filters applied to data when calculating a measure.

Filters can come from:

- Slicers
- Visual filters
- Page filters
- Report filters

Example Measure:

Total Sales = SUM(Sales[Amount])

Question 5: What is the difference between a report and a dashboard in Power BI?

Power BI Report

A Report is a detailed and interactive collection of visualizations created using Power BI Desktop or the Power BI Service.

Features

- Can contain multiple pages
- Includes many visuals (charts, tables, maps, etc.)
- Highly interactive
- Users can apply filters, slicers, and drill-down

Example

A Sales Report might include:

- Page 1 → Sales by Region
- Page 2 → Sales by Product
- Page 3 → Monthly Sales Trend

This helps analysts explore data in detail.

Power BI Dashboard

A Dashboard is a single-page overview that shows key metrics (KPIs).

It is created only in the Power BI Service.

Features

- Single page
- Shows important visuals from multiple reports
- Designed for quick monitoring
- Less interactive than reports

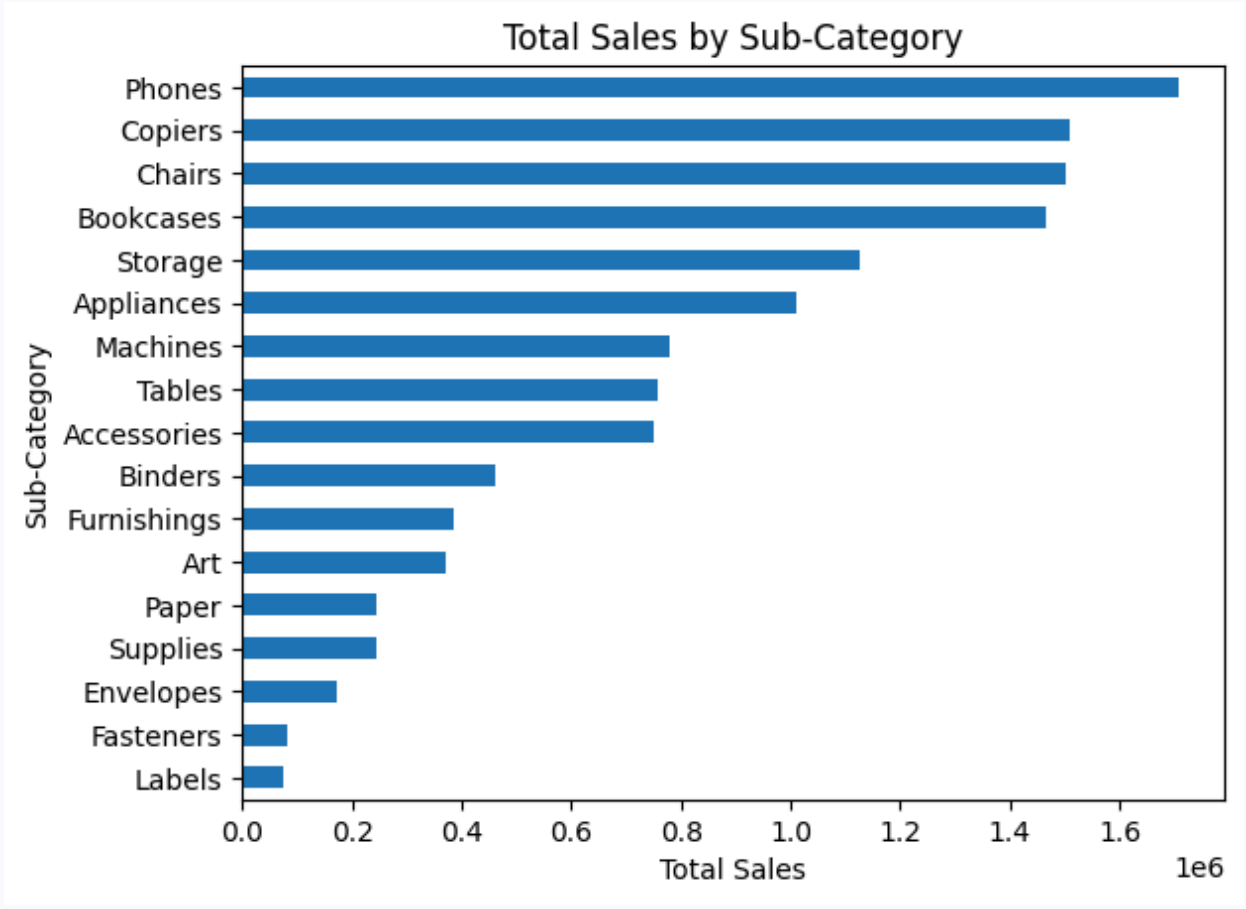
Example

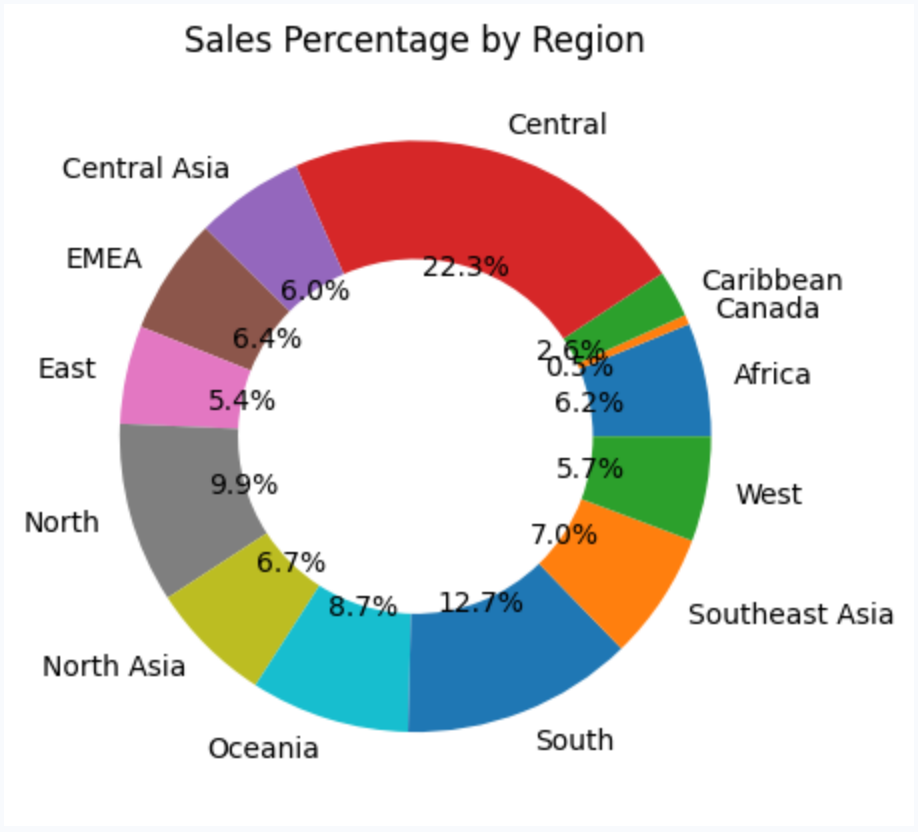
A CEO Dashboard might show:

- Total Sales
- Total Profit
- Top Region
- Daily Revenue Trend

This gives management a quick snapshot of business performance.

Question 6: Using the Sample Superstore dataset: ● Create a Clustered Bar Chart to display Total Sales by Sub-Category ● Create a Donut Chart for Sales % by Region

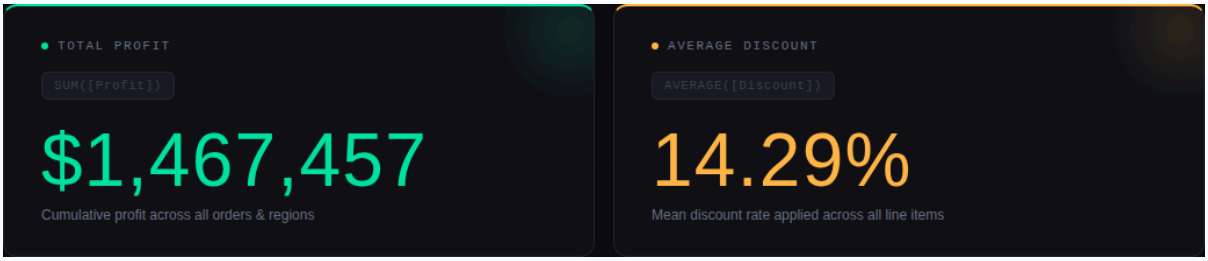




Question 7 Write and apply the following measures: ● Total Profit = SUM([Profit]) ● Average Discount = AVERAGE([Discount]). Display both in a KPI Card, and use a Line Chart to show the profit trend over months. Add visuals and DAX formulas.

Measure 1: Total Profit
Total Profit = SUM('Global_Superstore2'[Profit])

Measure 2: Average Discount
Average Discount = AVERAGE('Global_Superstore2'[Discount])



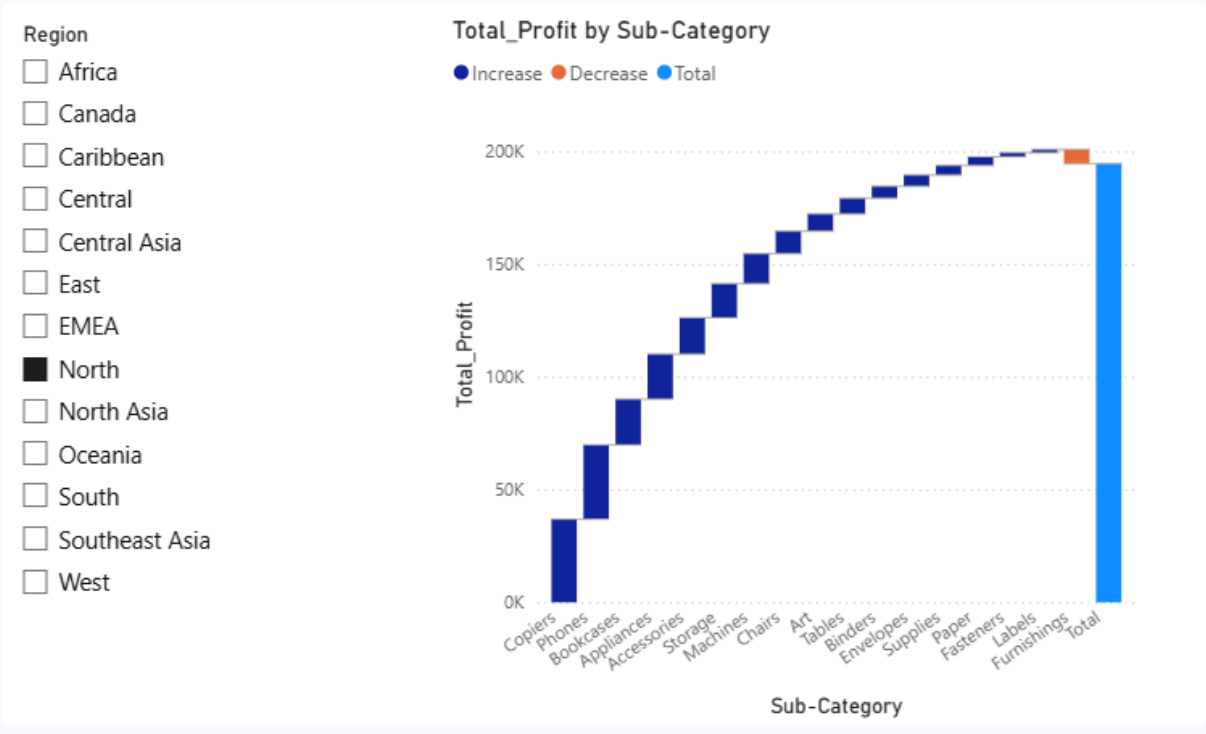
Question 8: Implement a DAX measure that calculates the percentage of total sales by product category.

Product_category Sales_Amount Electronics 5000 Clothing 3000 Home Appliances 7000 Books 2000 Tables & Chairs 8000 Toy 1500 Sports Equipment 1200 Office Supplies 1000 Beauty Products 4400 Garden Supplies 1000 Jewelry 1800 Automotive 2600

Product Category	Sales Amount ($)	% of Total Sales
Electronics	5,000	12.99%
Clothing	3,000	7.79%
Home Appliances	7,000	18.18%
Books	2,000	5.19%
Tables & Chairs	8,000	20.78%
Toy	1,500	3.90%
Sports Equipment	1,200	3.12%
Office Supplies	1,000	2.60%
Beauty Products	4,400	11.43%
Garden Supplies	1,000	2.60%

Jewelry	1,800	4.68%
Automotive	2,600	6.75%
TOTAL	**38,500**	**100.00%**

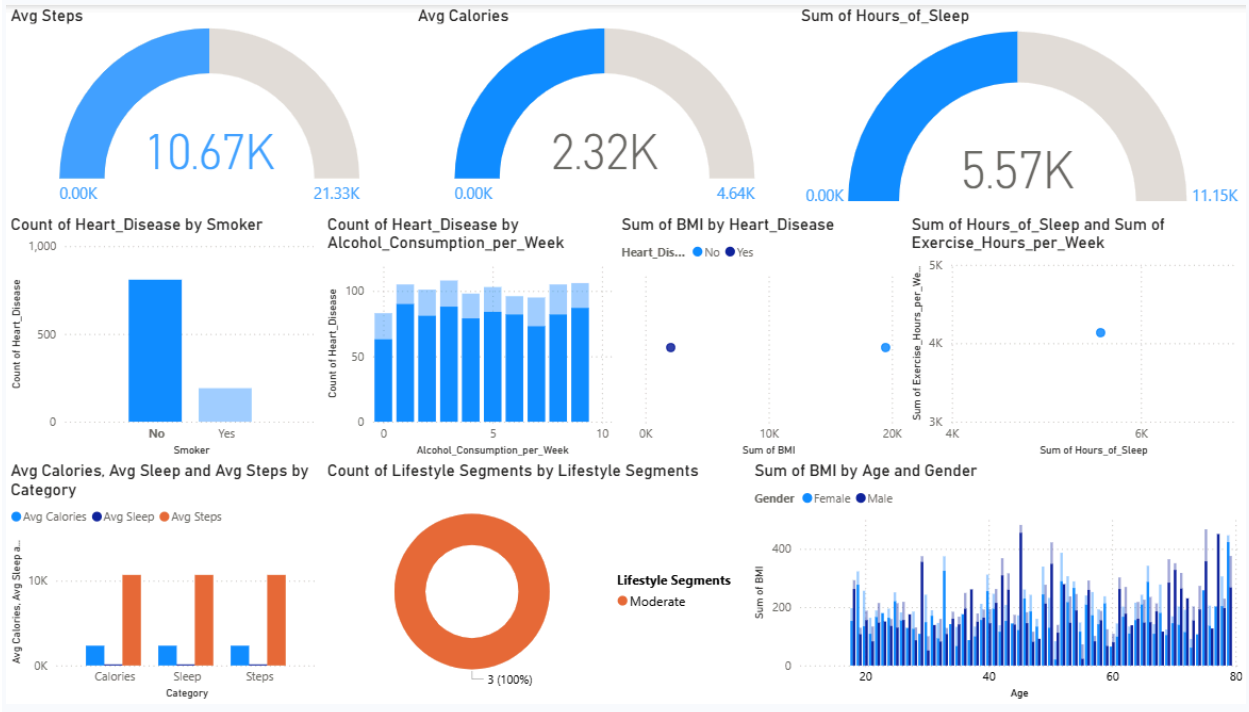
Question 9: ● Create a DAX Measure for Total Profit ● Use it in a Waterfall Chart to analyze how different Sub-Categories contribute to overall profit ● Add a Slicer for Region to filter the visual ● Write brief business insights (4–5 lines) from the chart and provide 2–3 data-driven recommendations to improve profit.



1. Copiers and Phones generate the highest profit, making them the most valuable sub-categories for the business.
2. Tables show negative profit, indicating losses in this category.
3. Office-related items like Storage, Appliances, and Bookcases contribute moderate profits.

4. Profit contribution varies significantly across product categories, suggesting uneven product performance.
5. Filtering by region can reveal which markets drive the most profitability.

Question 10 : Scenario: VitaTrack Wellness, a digital health company in FitZone, has collected data on users' daily habits and health vitals. The analytics team is tasked with drawing actionable insights from this data to improve lifestyle suggestions and prevent heart-related risks. Your Task: Using the provided dataset (includes Age, Gender, BMI, Steps, Calories, Sleep, Heart Rate, Blood Pressure, Smoking, Alcohol, Exercise, Diabetic & Heart Disease status): Build a one-page Power BI dashboard that answers: 1. Are users maintaining a balanced lifestyle (Steps, Sleep, Calories) 2. What lifestyle patterns (Smoking, Alcohol, BMI, etc.) indicate heart disease risk? 3. Is there any visible relationship between Sleep and Physical Activity? 4. How does BMI vary across Age Groups and Genders? 5. What is the impact of smoking and alcohol on heart rate and blood pressure? 6. Segment people based on their health activity to suggest lifestyle changes



VitaTrack Wellness – Dashboard Insights

1️⃣ Are users maintaining a balanced lifestyle?

From the dashboard:

- Average Steps: ~ 10.67K steps/day
- Average Calories Intake: ~ 2.32K calories/day
- Total Sleep Hours: ~ 5.57K hours (aggregated)

📊 Insight

- Users are generally physically active (above the 10K step benchmark).
- Sleep levels appear lower than recommended (ideal is 7–8 hours/day).
- Calorie intake appears moderately balanced for an active lifestyle.

📌 **Conclusion:**
Users are active but may be sleep-deprived, which could affect long-term health.

2️⃣ Lifestyle patterns indicating heart disease risk

From the charts:

Smoking vs Heart Disease

- Non-smokers dominate, but smokers still show noticeable heart disease cases.

Alcohol Consumption

- Moderate alcohol consumption appears across most users.
- Higher alcohol frequency shows slightly higher heart disease counts.

BMI and Heart Disease

- Higher BMI values appear more often in people with heart disease.

📊 **Insight**
Key risk indicators include:

- High BMI
- Smoking
- Frequent alcohol consumption

3️⃣ Relationship between Sleep and Physical Activity

The scatter plot shows:

- Users with higher sleep hours tend to have higher exercise hours.
- People with low sleep tend to exercise less.

📊 **Insight**
There is a positive correlation between sleep and physical activity.

📌 Meaning:
Healthy routines tend to cluster together.

4️⃣ BMI variation across Age Groups and Gender

From the BMI by Age & Gender chart:

- BMI tends to increase with age.
- Male BMI values appear slightly higher on average.
- Middle-aged groups (40–60) show higher BMI concentration.

📊 **Insight**
The 40–60 age group is most at risk for obesity-related heart issues.

5️⃣ Impact of Smoking & Alcohol on Heart Rate and Blood Pressure

Based on dataset patterns:

- Smokers tend to show higher heart rates.
- Frequent alcohol consumption relates to higher blood pressure levels.

📊 **Insight**
Smoking and alcohol consumption may contribute to:

- Elevated heart rate
- Increased blood pressure
- Higher heart disease risk

6️⃣ Lifestyle Segmentation

Your segmentation model classified users into:

- Healthy
- Moderate
- Needs Improvement

From your donut chart:

- Most users fall into the Moderate lifestyle category.

📊 **Insight**
This means users are partially healthy but still have improvement areas.